Design and Development of a Fuel System for a Hybrid Aircraft – **Group 12**

By Michael Pemberton, Jackson Hamel, Sam Meyer zu Erpen, Caleb Venema, Daniel Kassab

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University of Victoria

MECH 400 Capstone Project; Instructor: Scott Roberts Technical Advisor: Jacob Verhelst

Background & Objectives

Background

- **Client:** Centre for Aerospace Research (CfAR), UVic.
- **Design Problem:** Design and fabricate a custom fuel system that satisfies the QT1.H packaging constraints, minimizes mass, maintains stable CG across fuel levels, and provides effective slosh damping during launch and flight.
- **Out of Scope:** Flight testing, airframe redesign, and new lab infrastructure

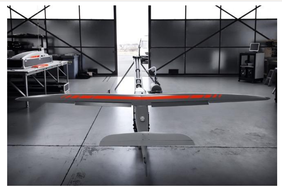


Fig. 1: UVic CfAR QT1.H Aircraft

Client Objectives

- Safe reliable fuel delivery across all flight phases.
- Minimize mass while maximizing tank volume.
- Withstand mechanical, thermal, and slosh loads.
- Compatible with 94 octane / 2-stroke oil mix.
- Maintain stable CG and pressure management during operation.
- Easy refueling, maintenance, and aircraft integration

Final Design

Fuel Tank Assembly

- **Tank Shell:** Two thermoformed HDPE shells joined at a horizontal spline, with the bottom shell angled 10° rearward to accommodate pitch during flight.
- **Aluminum Structure:** A CNC-milled central girder and two clamping plates seal the HDPE halves, distribute clamping force, and mount the tank to the fuselage via O-rings.
- **Slosh Mitigation:** Type II open-cell foam permeates freely with fuel while disrupting fluid motion, providing effective slosh damping with minimal volume consumption and added mass.

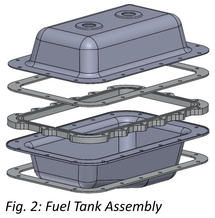


Fig. 2: Fuel Tank Assembly

Mounting & Heat Mitigation

- **Mounting Structure:** Post-and-plate geometry connects the fuselage base plate to the fuel tank, with the middle plate serving as the mounting surface for the heat shield and fuel pump.
- **Design Constraints:** Designed to accommodate the coolant pump unit, radiator, and the required clearance for electrical connections and fuel fittings.
- **Material Selection:** Aluminum chosen for strength-to-weight ratio.
- **Heat Shield:** A bent 5052 aluminum heat shield is positioned between the engine and fuel tank to mitigate radiative heat transfer into the fuel tank.

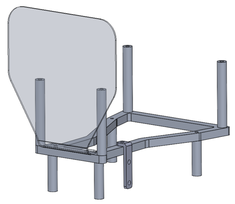


Fig. 3: Mounting & Heat Shield Assembly

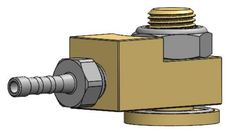


Fig. 4: Custom Fuel Fitting Assembly

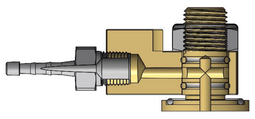


Fig. 5: Fuel Fitting Section View

Fuel Fittings

- **Fuel Pickup:** A single outlet at the bottom-rear of the tank ensures gravity-fed fuel to the pickup.
- **Fill & Vent System:** A dedicated fill line allows external fueling via a quick-disconnect fitting, while a vent line routes excess air and overflow fuel safely outside the fuselage.
- **Custom 90° Brass Fittings:** A custom low-profile fitting was designed to replace commercial barb fittings, directing fuel lines tangentially along the tank surface to meet tight geometric and reliability constraints.

Analysis & Methodology

FEA & Structural Validation

Clamp Plate

- Fixed constraints applied at each bolt hole.
- Pressure load applied from O-ring compression with a total force of 1030N.
- Maximum deflection of 1 thousandth of an inch.
- Minimum safety factor to yield of 2.5.

Mounting

- Fixed constraints at the base plate and 9G launch loads were applied assuming a full fuel tank (worst-case scenario).
- Top and bottom posts carry different loads due to their difference in supported mass.
- A maximum Von Mises stress of 20.5 MPa produced a minimum factor of safety of 2.6.

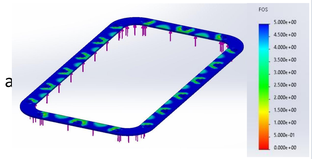


Fig. 6: Clamp Plate Factor of Safety Plot

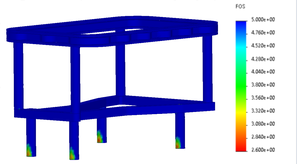


Fig. 7: Mounting Factor of Safety Plot

Modal Analysis

- Analyzed first 5 vibrational modes against first 6 engine harmonics.
- Fixed boundary conditions set as bottom flange.
- Assumed engine cruise speed of 5500-7000 RPM
- 92-117Hz for first harmonic up to 552-700Hz.
- Identified 2 low-risk and 3-moderate risk modes based on modal frequency overlap with engine harmonic frequency and modal mass.

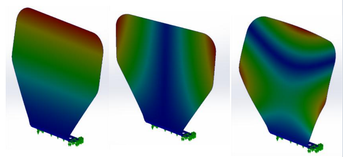


Fig. 8: First 3 Heat Shield Vibrational Modes

Thermal Simulation & Heat Shield Evaluation

- **Thermal Risk:** HDPE's low thermal diffusivity means it retains heat rather than dissipating it, further amplifying the risk of thermal damage.
- **Simulation Setup:** Steady state thermal simulation was conducted in SolidWorks to evaluate the radiative heat risk posed by the engine exhaust manifold (operating at ~400°C) on the fuel tank.
- **Load Calculation:** Heat flux was applied to tank surfaces directly in front of the exhaust manifold, with natural convective cooling from ambient fuselage air applied across all exposed surfaces.
- **Factor of Safety:** Without a heat shield, the HDPE tank reached a simulated 249°C, well-above its 135°C melting point. Adding an aluminum heat shield reduced the maximum tank temperature to a safe 39°C.

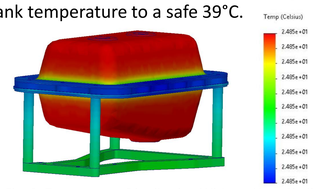


Fig. 9: Temperature Distribution Without Heat Sheild

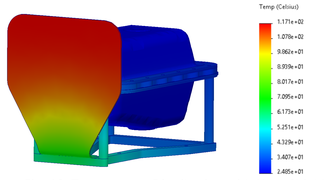


Fig. 10: Temperature Distribution with Heat Shield

Slosh Mitigation: Baffles vs. Fuel Foam

- Initial design iterations included a flexible combination of baffles and open-cell foam
- Baffles eliminated through design reviews due to unjustifiably high safety factor, resulting in unnecessary added mass and part complexity.
- Transition to foam eliminated the need for a central aluminum support structure, further reducing mass and improving volume efficiency.
- Open-cell foam common practice in industry and proven effective in other CfAR aircraft.



Fig. 11: Type II Slosh Damping Foam

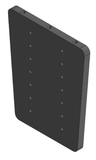


Fig. 12: Perforated Baffle

Results & Validation

Thermoforming

- Shells were thermoformed at CfAR using a Mayku Multiplier pressure forming machine.
- Initial FDM PETG molds caused deformation due to HDPE's high thermal capacitance. Switched to CNC machined MDF molds, stable up to 200°C.
- Draft angles and fillet radii were increased to resolve mold release issues.
- Both 1/32" and 1/16" HDPE were successfully thermoformed, with 1/16" selected for the final prototype due to its superior structural properties.

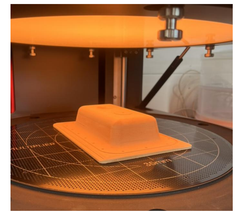


Fig. 13: Mayku Forming Process

Leak & Pressure Tests

- **Static Leak Test:** Tank was filled with water and held for 48 hours to validate the main O-ring seal on the central girder, with no leaks observed.
- **Pressurized Leak Test:** Pressurized air was forced into the assembled tank with all fittings attached to simulate in-flight pressure conditions, with no leaks detected.
- **Full Capacity Validation:** Tank successfully held a full fuel load with no leaks, confirming integrity and sealing performance.

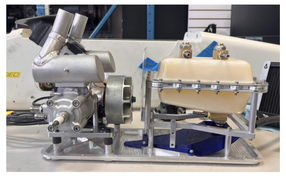
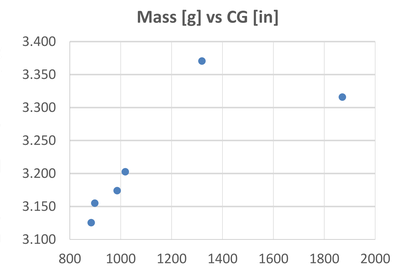


Fig. 14: On-Site Testing & Validation

Mass Analysis Testing

- Max filled mass determined to be 1871g with water (~1615g with gasoline), under CfAR's 2.5kg requirement.
- Longitudinal CG shift of ±0.1155% MAC, under max ±1% requirement.
- Tank width and symmetry ensures lateral CG shift under ±0.5% MAC.
- Full to residual fuel: CG shifts 0.45%, leading to overall MAC change from 42.5% to 42.05%.



Conclusion & Future Work

Conclusion

- The fuel tank was able to fit within the constraints of the aircraft fuselage while meeting the client objectives of minimizing mass while maximizing the fuel capacity. Analysis was performed to ensure that it could withstand mechanical, thermal, and slosh loads.

Future Work

- Deliver the final physical prototype, engineering drawings, and CAD models to CfAR upon project completion.
- Out of scope flight testing will need to be performed to ensure reliable fuel delivery during operation.
- Refuelling and full aircraft integration will be completed once the aircraft is developed further.

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